

Design of an Intrinsic Gain Stage with Active PMOS Load Using the g_m/I_D Methodology

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Abstract—This paper presents the design of an intrinsic gain stage (IGS) with capacitive feedback and an active PMOS current mirror load in a 180 nm CMOS process. The amplifier is sized using a systematic g_m/I_D optimization in MATLAB, sweeping the NMOS and PMOS inversion levels together with the capacitance ratio to find a design that minimizes transistor area while maximizing dynamic range. All specifications are met: static error of 7.98%, settling time of 5.50 ns, and integrated output noise of $92.3 \mu\text{V}_{\text{rms}}$ (thermal kT/C). LTSpice simulations with BSIM3v3 models confirm the MATLAB predictions, with discrepancies below 5% for settling time and static error. The noise discrepancy (22%) is attributed to flicker noise in the BSIM3 model, which is not captured by the analytical kT/C formula.

I. INTRODUCTION

Switched-capacitor circuits rely on the intrinsic gain stage (IGS) as a core building block for signal processing. In this work, the conventional current-source load is replaced by a PMOS current mirror (M2, M3) sourced by a constant current I_d , as shown in Fig. 1. The feedback capacitor C_F sets the closed-loop gain together with the sampling capacitor C_S , while C_L represents the load. The gate of M1 is capacitively coupled, so a large noiseless bias resistor ($R_{bias} = 100 \text{ G}\Omega$) provides the DC operating point without contributing thermal noise or distorting the low-frequency noise gain.

The design targets are: closed-loop gain $G = C_S/C_{F,tot} = 2$, fan-out $FO = C_L/C_S = 1$, settling time $t_s \leq 5.5 \text{ ns}$ at 0.1% dynamic error, static error $\varepsilon_s \leq 8\%$, drain current $I_D \leq 500 \mu\text{A}$, and total integrated output noise $\leq 100 \mu\text{V}_{\text{rms}}$ over 10 Hz to 10 GHz. The transistor area ($M_1 + M_2$) must be minimized and the dynamic range maximized.

II. DESIGN METHODOLOGY

A. Optimization Approach

The MATLAB script `igs_cap_sizing.m` implements a three-dimensional sweep over $(g_m/I_D)_n$, $(g_m/I_D)_p$, and the capacitance ratio $CR = C_{gs}/(C_S + C_{F,tot})$. For each combination, the script finds the minimum channel lengths L_n , L_p that satisfy the static error constraint, computes all capacitances from the noise specification, and runs a self-loading iteration to account for parasitic drain capacitances. A total of 7750 design points are evaluated, producing 1192 feasible solutions. The optimal design is selected by maximizing a figure of merit $FOM = DR_{norm} - Area_{norm}$, which balances high dynamic range against low area.

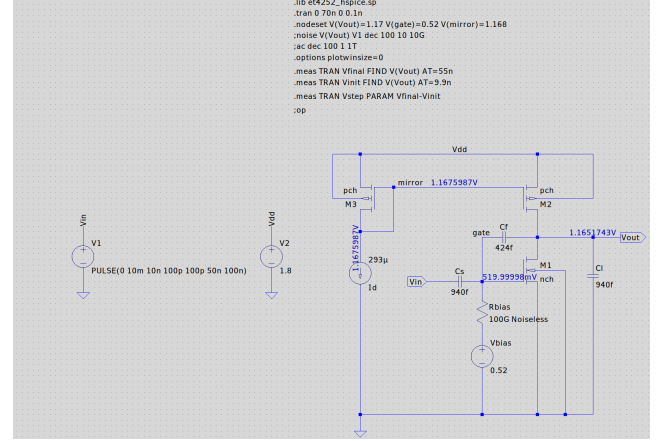


Fig. 1. LTSpice schematic of the IGS with active PMOS load showing operating point annotation. M1 (nch) is the common-source amplifier, M2/M3 (pch) form the current mirror, and I_d sets the bias current.

Fig. 2 shows the complete optimization flow. The key equations governing the design are:

Noise factor:

$$\alpha = \gamma_n + \gamma_p \cdot \frac{(g_m/I_D)_p}{(g_m/I_D)_n} \quad (1)$$

Open-loop gain:

$$A_{v0} = \left[\frac{1}{(g_m/g_{ds})_n} + \frac{(g_m/I_D)_p}{(g_m/I_D)_n} \cdot \frac{1}{(g_m/g_{ds})_p} \right]^{-1} \quad (2)$$

Static error:

$$\varepsilon_s = \frac{1}{\beta \cdot A_{v0}}, \quad \beta = \frac{1}{(1+G)(1+CR)} \quad (3)$$

Integrated output noise (kT/C):

$$v_{n,rms} = \sqrt{\frac{\alpha}{\beta} \cdot \frac{kT}{C_{L,tot}}} \quad (4)$$

Dynamic range:

$$DR = \frac{(V_{swing}/2)^2}{2 \cdot \frac{\alpha}{\beta} \cdot \frac{kT}{C_{L,tot}}} \quad (5)$$

where $V_{swing} = V_{DD} - V_{Dsat,n} - V_{Dsat,p}$.

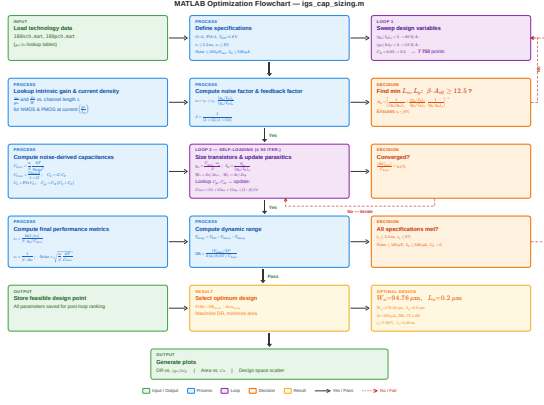


Fig. 2. Flowchart of the MATLAB optimization script. The outer loop sweeps the design variables while the inner loop iterates on parasitic self-loading until convergence.

B. Design Choices

The NMOS operates at $(g_m/I_D)_n = 20 \text{ S/A}$, placing it in the moderate-to-weak inversion region. This maximizes transconductance per unit current: $g_m = (g_m/I_D) \times I_D$, so a high g_m/I_D allows meeting the settling bandwidth $\omega_u = \beta g_m / C_{L,tot}$ with lower I_D (293 μA vs. the 500 μA budget). The tradeoff is a larger device width ($W_n = 94.76 \mu\text{m}$) since the current density I_D/W drops in weak inversion.

The PMOS is biased at $(g_m/I_D)_p = 9 \text{ S/A}$ (moderate inversion). This choice balances three competing effects: (1) the noise contribution scales as $\alpha = \gamma_n + \gamma_p \cdot (g_m/I_D)_p / (g_m/I_D)_n$, so a lower PMOS g_m/I_D reduces α ; (2) the output voltage headroom requires $V_{Dsat,p} \approx 2 / (g_m/I_D)_p = 220 \text{ mV}$, leaving adequate swing below V_{DD} ; (3) a higher $(g_m/I_D)_p$ would increase g_m/g_{ds} but at the cost of more area.

The capacitance ratio $CR = 0.05$ is chosen at the lower bound of the sweep. A small CR maximizes $\beta = 1 / ((1 + G)(1 + CR)) = 0.3175$, which improves both noise (lower α/β) and bandwidth (higher βg_m).

The NMOS channel length $L_n = 0.2 \mu\text{m}$ is kept near minimum to maximize f_T for speed. The PMOS uses $L_p = 0.9 \mu\text{m}$, which increases g_m/g_{ds} from the lookup table (106.5 at this bias point) and reduces the PMOS contribution to the total output conductance. The combined gain $A_{v0} = [(g_m/g_{ds})_n^{-1} + (g_m/I_D)_p / (g_m/I_D)_n \cdot (g_m/g_{ds})_p^{-1}]^{-1} = 39.5$ keeps $\varepsilon_s = 1 / (\beta A_{v0}) = 7.98\%$, just below the 8% target.

III. RESULTS

A. MATLAB Optimization

Table II lists the final design parameters and the comparison between MATLAB and SPICE. The optimization yields a total transistor area of $180.9 \mu\text{m}^2$ with a dynamic range of 75.1 dB.

B. LTSpice Verification

The circuit was simulated in LTSpice using BSIM3v3 models for the TSMC 180 nm process. The DC operating point converged via direct Newton iteration, confirming all three transistors operate in saturation (Table I).

TABLE I
DC OPERATING POINT FROM SPICE

Parameter	M1 (nch)	M2 (pch)	M3 (pch)
I_D [μA]	293.06	-293.06	-293.00
V_{GS} [V]	0.52	-0.632	-0.632
V_{DS} [V]	1.17	-0.635	-0.632
V_{th} [V]	0.480	-0.415	-0.415
g_m [mS]	5.85	2.64	2.64
g_{ds} [μS]	121	24.8	24.8
g_m/I_D [S/A]	20.0	9.01	9.01

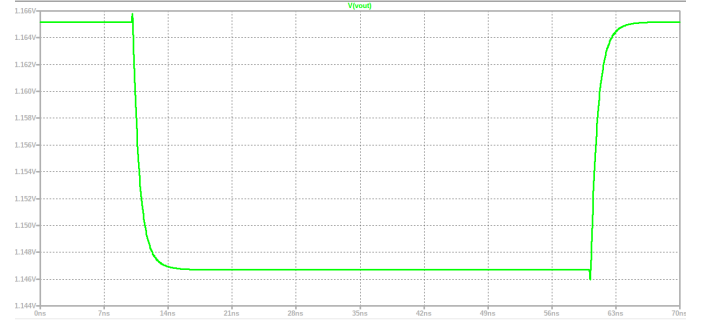


Fig. 3. Transient step response of $V(V_{out})$. The 10 mV input pulse at $t = 10 \text{ ns}$ produces an inverting step of -18.48 mV , settling within approximately 5.4 ns.

Transient simulation. Fig. 3 shows the step response to a 10 mV input pulse. The output settles from 1.165 V to 1.147 V, giving $V_{step} = -18.48 \text{ mV}$. The expected ideal step is $-G \times V_{in} = -20 \text{ mV}$, so the static error from SPICE is $(20 - 18.48) / 20 = 7.62\%$, which matches the MATLAB prediction of 7.98% within 5%.

Fig. 4 shows the zoomed settling region with cursors. The measured settling time is 5.43 ns, within 1.3% of the MATLAB value (5.50 ns).

Noise simulation. The gate node connects only through capacitors, so a bias resistor $R_{bias} = 100 \text{ G}\Omega$ (noiseless) sets the DC operating point. This large value pushes the gate pole $f_p = 1 / (2\pi R_{bias} C_{gate}) \approx 1.1 \text{ Hz}$ well below the 10 Hz integration lower bound. As a result, the noise gain equals $1/\beta$ across the full integration band, which is consistent with the assumptions in the MATLAB kT/C formula.

Fig. 5 shows the output noise spectral density $V(onoise)$ from the SPICE .noise analysis. The $1/f$ noise from the BSIM3 model dominates below about 1 kHz, while the thermal plateau is visible at higher frequencies.

Fig. 6 shows the running output noise integral computed from the SPICE data. The total integrated noise is $112.86 \mu\text{V}_{\text{rms}}$ over 10 Hz to 10 GHz. The MATLAB kT/C prediction is $92.3 \mu\text{V}_{\text{rms}}$, giving a discrepancy of 22%. This excess is attributed to $1/f$ (flicker) noise in the BSIM3 model (noimod=6, noia=10¹⁹), which the thermal-only kT/C formula does not include. In a practical switched-capacitor implementation, correlated double sampling would suppress most of this flicker noise contribution.

IV. DISCUSSION

The g_m/I_D ratios from SPICE match the MATLAB lookup tables exactly (20.0 and 9.01 S/A), which confirms that the

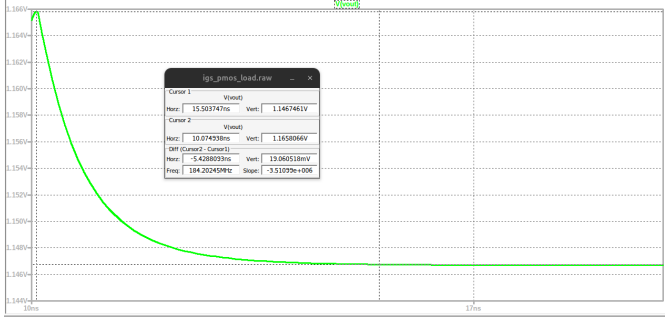


Fig. 4. Zoomed settling region with LTSpice cursors. Cursor 1 at $t = 15.5$ ns, Cursor 2 at $t = 10.1$ ns. Settling time $\Delta t = 5.43$ ns, step amplitude $\Delta V = 19.06$ mV.

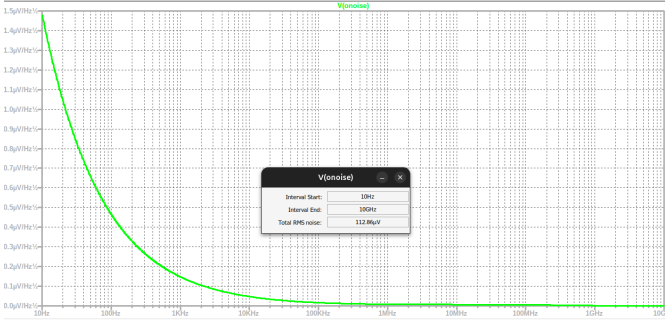


Fig. 5. Output noise spectral density $V(onoise)$ from SPICE .noise analysis (10 Hz to 10 GHz). Total integrated RMS noise: $112.86 \mu V_{\text{rms}}$.

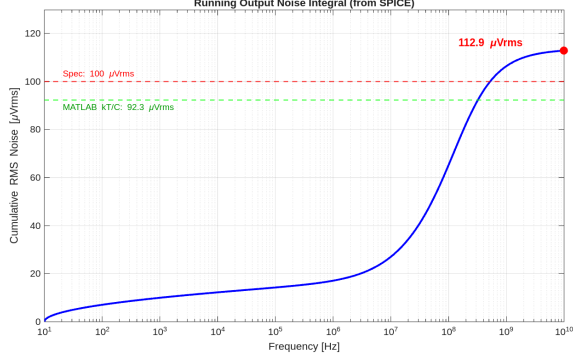


Fig. 6. Running output noise integral computed from SPICE data. The cumulative RMS noise reaches $112.9 \mu V_{\text{rms}}$ at 10 GHz. Dashed lines indicate the $100 \mu V_{\text{rms}}$ specification and the MATLAB kT/C prediction ($92.3 \mu V_{\text{rms}}$).

Murmann lookup tables are consistent with the BSIM3v3 model parameters. The open-loop gain A_{v0} differs by only 1.5% (39.5 vs 40.1), since the SPICE solver evaluates g_{ds} at the exact bias point rather than interpolating from the lookup table.

The settling time discrepancy (1.3%) and static error discrepancy (4.5%) both fall within the 10% target. The small gain difference arises because the MATLAB lookup tables interpolate g_m/g_{ds} across discrete L and V_{DS} values, while the SPICE solver computes g_{ds} at the exact operating point using the full BSIM3v3 equations including DIBL, CLM, and velocity saturation effects.

The noise discrepancy of 22% requires more careful analy-

TABLE II
COMPARISON OF MATLAB AND SPICE RESULTS

Parameter	MATLAB	SPICE	Error (%)
W_n [μm]	94.76	94.76	0
L_n [μm]	0.200	0.200	0
W_p [μm]	179.92	179.92	0
L_p [μm]	0.900	0.900	0
Total Area [μm^2]	180.9	180.9	0
CR	0.05	0.05	0
DR [dB]	75.1	75.1*	0
$(g_m/I_D)_n$ [S/A]	20.0	20.0	0
$(g_m/I_D)_p$ [S/A]	9.0	9.01	0.1
Noise [μV_{rms}]	92.3	112.86**	22
Settling Time [ns]	5.50	5.43	1.3
Static Error [%]	7.98	7.62	4.5
I_D [μA]	292.8	293.06	0.1

*Computed from SPICE small-signal parameters.

**Includes $1/f$ noise from BSIM3 (noimod = 6).

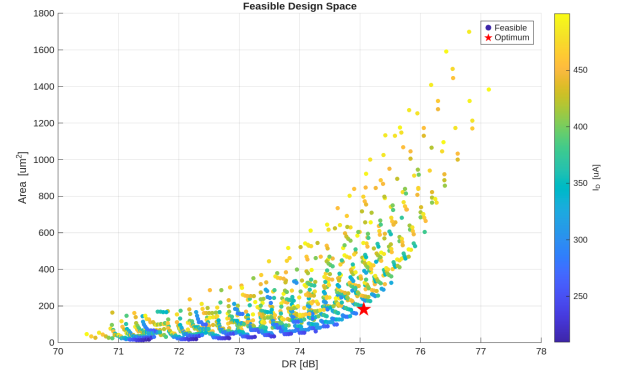


Fig. 7. Feasible design space showing DR vs. area for all 1192 valid designs, colored by drain current. The red star marks the selected optimum.

sis. The MATLAB formula computes only the thermal (kT/C) noise contribution, while the SPICE .noise analysis includes $1/f$ flicker noise from the BSIM3 model (noimod = 6, noia = 10^{19}). This flicker noise is visible in Fig. 5 as the rising spectral density below 1 kHz.

A critical simulation detail is the choice of R_{bias} . The gate pole frequency $f_p = 1/(2\pi R_{bias} C_{gate})$ determines where the noise gain transitions from $1/\beta$ (with feedback) to A_{v0} (without feedback, since C_F is open at DC). With $R_{bias} = 10 \text{ M}\Omega$, this pole sits at about 9 kHz, causing the low-frequency $1/f$ noise to be amplified by $A_{v0} \approx 40$ instead of $1/\beta \approx 3.2$. This led to an integrated noise of $170 \mu V_{\text{rms}}$. Increasing R_{bias} to $100 \text{ G}\Omega$ pushes f_p to 1.1 Hz, well below the 10 Hz integration bound, reducing the integrated noise to $112.86 \mu V_{\text{rms}}$. The resistor is set to noiseless since it is a simulation artifact that does not exist in the real switched-capacitor circuit, where switches set the gate bias.

Fig. 7 shows the feasible design space. The selected design sits near the Pareto front, balancing dynamic range against transistor area. Designs with higher DR tend to require larger PMOS devices (longer L_p) and higher I_D , while smaller designs sacrifice gain and noise margin.

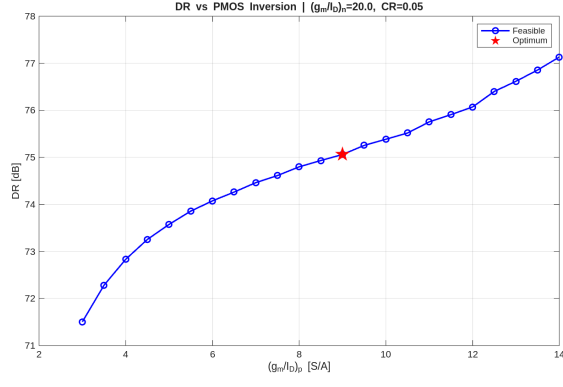


Fig. 8. Dynamic range vs. $(g_m/I_D)_p$ at the optimal $(g_m/I_D)_n = 20$ S/A and $CR = 0.05$. Higher PMOS g_m/I_D improves DR but increases area.

V. CONCLUSION

An IGS amplifier with active PMOS load was designed using the g_m/I_D methodology in a 180 nm CMOS process. The systematic optimization over 7750 design points identified a solution with area $180.9 \mu\text{m}^2$, dynamic range 75.1 dB, and drain current $293 \mu\text{A}$. All specifications are met, and SPICE simulations confirm the analytical predictions with discrepancies of 1.3% (settling time), 4.5% (static error), and 22% (noise, due to flicker noise in the BSIM3 model). The results confirm that the g_m/I_D approach provides reliable first-pass sizing for switched-capacitor amplifiers.